

soils4med

SOIL health monitoring and information systems FOR
sustainable soil management in the MEDiterranean region

MedSIS manual

Administrators

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DATASETS



MAPS



DOCUMENTS



GEOSTORIES



DASHBOARDS

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Delivery dates: **20 Settembre 2026**

Table of contents:

1. Overview	4
2. Installation	5
3. Management	7
4. Backup and Restore	16

1. Overview

A Soil Information System (SIS) is an integrated information system designed to facilitate the storage, analysis, management and dissemination of soil data.

To assess and monitor soil health and promote sustainable soil management in the Mediterranean region, there is a pressing need to improve the availability and accessibility of soil data.

One of the objectives of the SOILS4MED project (SOIL health monitoring and information systems FOR sustainable soil management in the MEDiterranean region) which addresses this need by designing and developing a decentralised network of country-based soil information systems.

This network aims to improve the interoperability, management, governance, and utilization of soil data across the Mediterranean region by establishing a federated architecture of SISs, starting with the development of a harmonized data model aligned with the World Reference Base for Soil Resources (WRB 2022), alongside the implementation of core functionalities for the ingestion, integration, and management of both legacy and newly acquired soil data.

The SIS is essentially a **Web GIS Catalogue** of resources related to soil data. It is based on Geonode (<https://geonode.org/>), whose resources include datasets, maps, documents, geostories and dashboards. In addition to the standard catalogue functionalities for managing **soil data**, MED-SIS includes an additional tool, known as the **Back Office**

MED-SIS staff use the Back Office to create and configure catalogue resources and to manage them through the Web GIS catalogue.

The **Med-SIS** was designed and implemented by **CRS4** (<https://www.crs4.it/>) (Roberto Demontis, Eva Lorrai, Laura Muscas, Piergiorgio Palla) with the support of all project partners.

The code is located in a GitHub repository: https://github.com/crs4/med_sis licensed under the GNU General Public License v3.0 (GPL-3.0) see LICENSE file for the full text.

Further details on Web GIS Catalogue functionalities can be found on the official GeoNode documentation page at <https://docs.geonode.org/en/latest>.

Video and manuals are accessible in the “Help” page of the MED-SIS or in the Github repository.

Soils4Med is a project funded by **PRIMA** (<https://prima-med.org/>) - Partnership for Research and Innovation in the Mediterranean Area.

This manual has been created as a reference guide to assist Med-SIS administrators with system installation and management, and is divided into four main sections: "Installation," "Management," "Administration Panel," and "Data Backup and Recovery."

2. Installation

The main components of the Med-SIS are two:

Component	Role	Stack
Web GIS Catalogue	Soil data management	Geonode (https://geonode.org)
Back Office	Web utilities for Soil data management	Nextjs (https://nextjs.org)

The **Web GIS Catalogue** is a Geonode WEB GIS catalogue for Soil data management

Main libraries:

- Geonode 5.1.0
- Geoserver 2.28.4
- Django 5.2.17
- Django Rest Framework 3.17.2
- Python 3.12

The **Back Office** is a NextJS application that use the PrimeReact REACT Diamond template

Main libraries:

- React 18.3.0
- NextJS 15.5.15
- Primereact 10.2.1

Getting started

Med-SIS is a Docker Compose multi container application tested in a Ubuntu server 24.04

The minimal requirements are:

- Operating system:** Ubuntu server 24.04
- Storage :** 1 TB
- CPU:** Intel 64-bit, 4.0 GHZ, 8 cores, 2MB Cache
- The **RAM:** 32 GB
- Software:** Docker Compose (Desktop or Engine)

Docker Compose is a tool for defining and running multi-container applications. It is the key to unlocking a streamlined and efficient development and deployment experience.

Compose simplifies the control of your entire application stack, making it easy to manage services, networks, and volumes in a single YAML configuration file. Then, with a single command, you create and start all the services from your configuration file.

The link <https://docs.geonode.org/en/latest/setup/docker/prerequisites/> might be useful for installing Docker.

Step 1 clone the repository

```
```bash
 git clone https://github.com/crs4/med_sis.git
 cd med_sis
```
```

Step 2 copy the .env.sample file in a new file named .env

```
```bash
 cp .env.sample .env
```
```

Step 3 edit the environment variables in the new .env file

Use a system editor and consult the comments in the file.

Step 4 run docker compose build

```
```bash
 docker compose build
```
```

Step 5 run the docker compose application

```
```bash
 docker compose up -d
```
```

Step 5 access and test the site

Using the browser and the configured address and domain

3. Management

The Admin Panel permits configuring the geonode catalogue. Only staff user can access to the panel

Admin Panel Access:

- URL: `http://<host_name>/admin`
- Menu: Via the "Admin" link in the top user menu.
- Permissions: Requires the "staff" or "superuser" flag in the user profile.

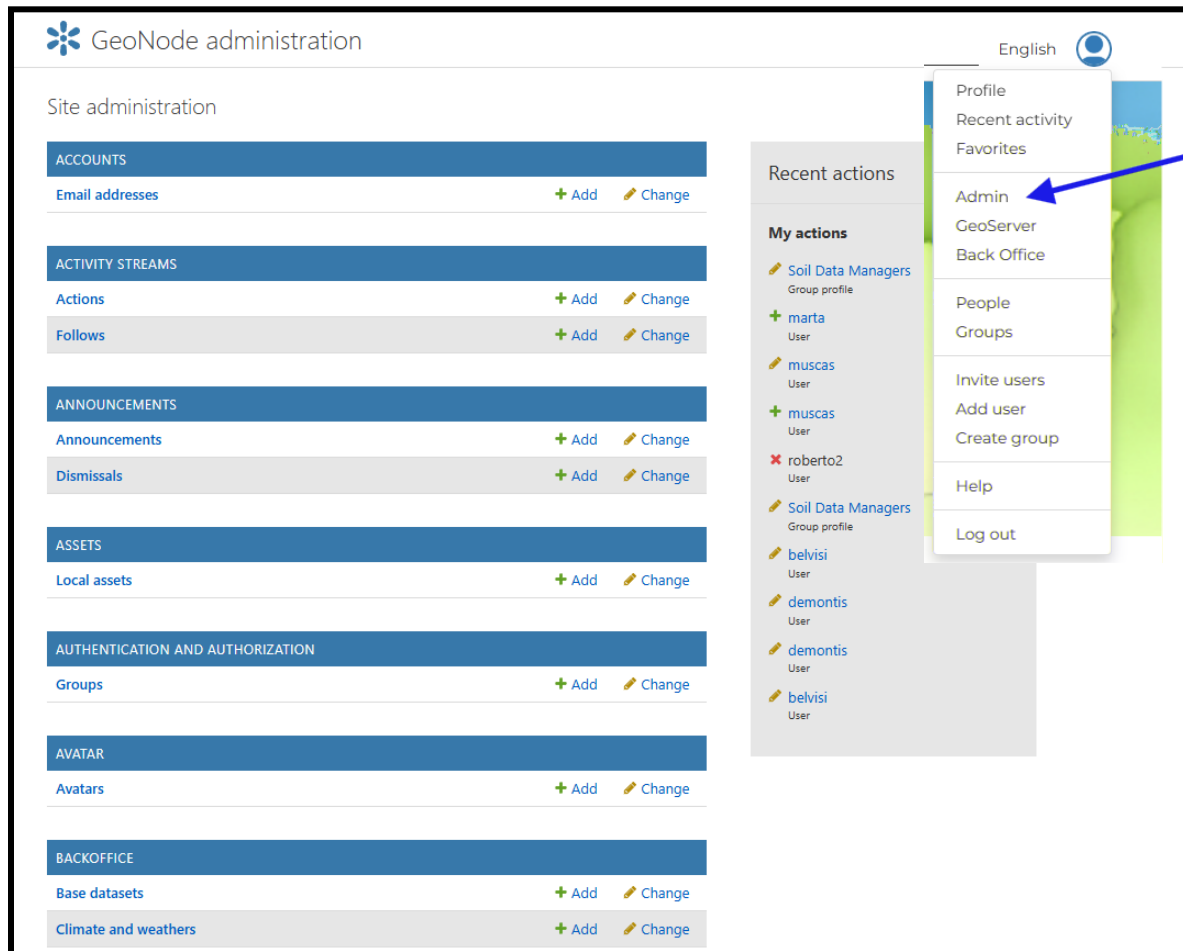


Figure 1 The Admin panel

see the GeoNode documentation at: <https://docs.geonode.org/projects/v4/en/4.4.x/admin/index.html>

USERS

The user can be managed using the Admin Panel or in the “People” page of the Catalogue.

Med-SIS staff have the necessary permissions to open the "People" or "Groups" pages (Figure 3) using the user menu at the top of the page.

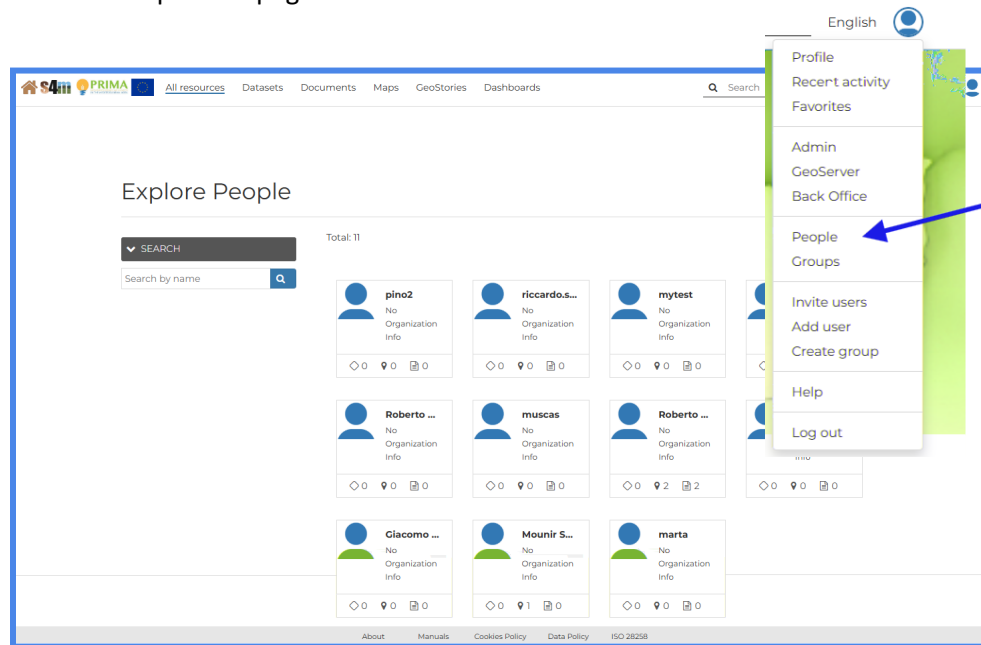


Figure 2 The People page

Clicking on a user card opens the corresponding profile.

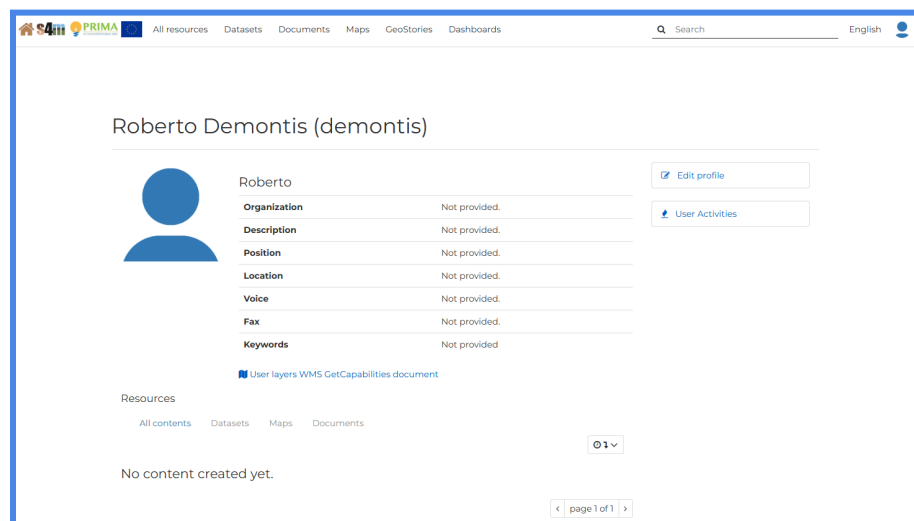


Figure 3 user profile

On the user profile page, you can edit the profile and view the user's activities and datasets.

The Admin panel permits to modify more other user properties.

Figure 4: Link to users list in the admin panel

Figure 5: List of users

The user list contains general information about registered users:

ID: the id of the user

USERNAME: username

ORGANIZATION NAME: organization name of the user

EMAIL ADDRESS: email address of the user

FIRST NAME: first name of the user

LAST NAME : last name of the user

STAFF STATUS : indicates whether or not the user is a staff member.

ACTIVE : indicates whether the user is enabled or not.

The Action drop down menu permits to delete the selected user or to set resource permission.

The button "Add User" permits to

The self registration is closed

(.env: REGISTRATION_OPEN=False)

The user can register in the system independently and is **active** after registration

(.env: ACCOUNT_APPROVAL_REQUIRED=False)

(.env: REGISTRATION_OPEN=True)

User registration must be approved by the MED-SIS staff and **isn't active** prior to approval

(.env: ACCOUNT_APPROVAL_REQUIRED=True)

(.env: REGISTRATION_OPEN=True)

A registered user can access all resources for which "View," "Download," "Edit," or "Manage" permissions have been set for the REGISTERED USERS group or for the user themselves or for a group of which the user is a member.

The screenshot shows a web interface for changing a user. At the top, it says 'Change user' and 'demonitis'. There are two buttons: 'HISTORY' and 'VIEW ON SITE'. Below this, there's a 'Username:' field with 'demonitis' and a note: 'It requires 1-30 characters or lower letters, digits and the /-+_! only.' Below that is a 'Password:' field with a 'Reset password' button and a note: 'Raw passwords are not stored, so there is no way to see the user's password.' Below the password field is a 'Personal info' section with fields for 'First name:' (Roberto), 'Last name:' (Demonitis), and 'Email address:' (demonitis.roberto@pippo.com). Below that is a 'Permissions' section with three checkboxes: 'Active' (checked), 'Staff status' (unchecked), and 'Superuser status' (unchecked). Each checkbox has a description: 'Designates whether this user should be treated as active. Unselect this instead of deleting accounts.', 'Designates whether the user can log into this admin site.', and 'Designates that this user has all permissions without explicitly assigning them.'

Figure 6: Change user - general info

The “Change User” shows a form that permits to reset password and manage the user status:

ACTIVE: the user is active and can access to the catalogue

STAFF STATUS: Designates whether the user can log into the admin panel

SUPERUSER STATUS: the user is an administrator with full permissions

Groups:

Available groups

Choose groups by selecting them and then select the "Choose" arrow button.

contributors

data-managers

Choose all groups

The groups this user belongs to. A user will get all permissions granted to each of their groups. Hold down "Control", or "Command" on a Mac, to select more than one.

Chosen groups

Remove groups by selecting them and then select the "Remove" arrow button.

anonymous

registered-members

Remove all groups

important dates

Last login:

Date: 2026-10-04 Today

Time: 17:54:32 Now

Note: You are 2 hours ahead of server time.

Date joined:

Date: 2026-08-12 Today

Time: 12:34:21 Now

Note: You are 2 hours ahead of server time.

Figure 7: Change user - groups setting

Also is possible to assign the user to groups (Figure 7) and change profile (Figure 8). The panel in the bottom of the page permits to save changes or delete the user.

Extended profile

Organization Name:

name of the responsible organization

Profile:

Introduce yourself

Position:

role or position of the responsible person

Voice:

telephone number by which individuals can speak to the responsible organization or individual

Facsimile:

telephone number of a facsimile machine for the responsible organization or individual

Delivery Point:

physical and email address at which the organization or individual may be contacted

City:

city of the location

Administrative Area:

state, province of the location

Postal Code:

ZIP or other postal code

SAVE

Save and add another

Save and continue editing

Delete

Figure 8: Change user - extended profile

GROUPS

The user can be managed using the Admin Panel or in the “Groups” page of the Catalogue.

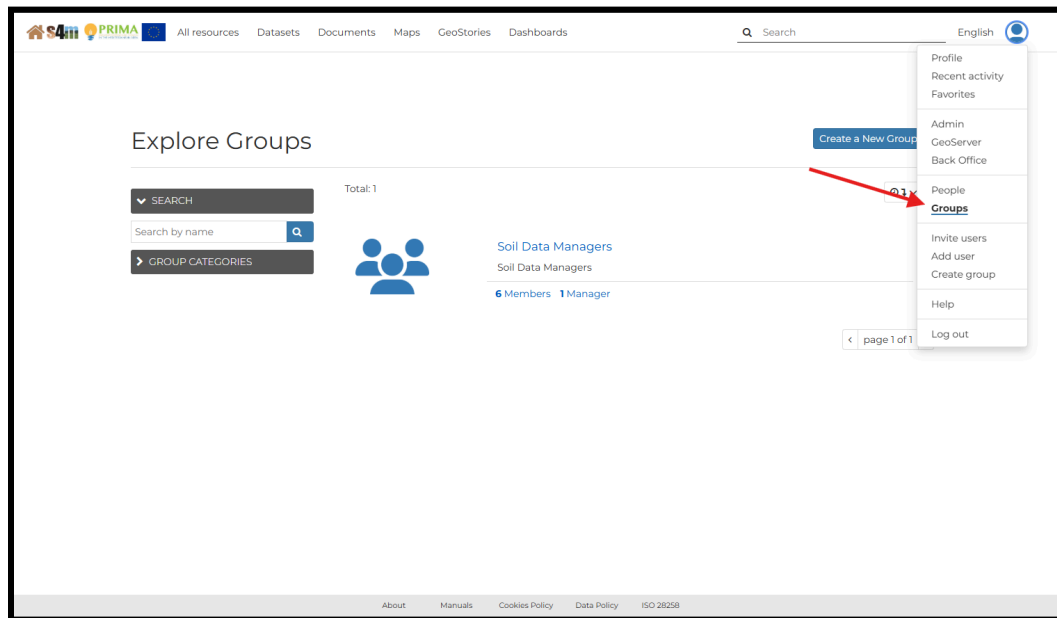


Figure 9: list of groups

Clicking on a groupcard opens the corresponding profile.

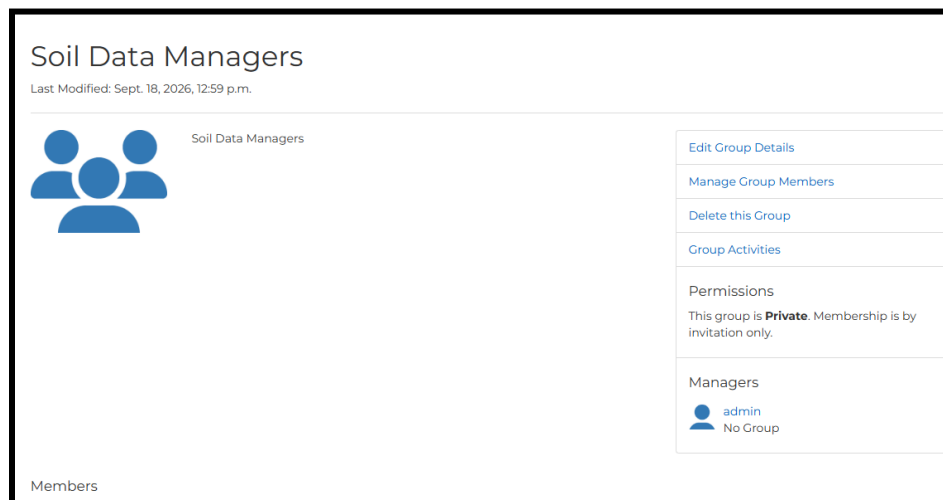


Figure 10: group description

The page in Figure 7 shows the members of the group and the managers. The panel on the right permits to perform some actions on group

Edit group details
Manage group members
Show group activities
Delete the group

Group **permissions** constitute the rule for member involvement. A group can be **private** or **public**.

Adding a member to a **private** group requires staff intervention.

All registered users can join a **public** group without moderation.

All group permissions are inherited by all users who are members of the group.

The Admin panel permits to modify more other group properties and permissions.

GeoNode administration

WELCOME, ROBERTO. [VIEW SITE](#) / [CHANGE PASSWORD](#) / [LOG OUT](#)

Home > Groups > Group profiles > Soil Data Managers

Sample processing tasks [Add](#)

GEONODE SECURITY

Authentication Configurations [Add](#)

URL Pattern Authentication Configurations [Add](#)

GEONODE THEMES LIBRARY

Jumbotron theme slides [Add](#)

Themes [Add](#)

GROUPS

Group Categories [Add](#)

Group profiles [Add](#)

HARVESTING

Asynchronous harvesting sessions

Harvestable resources [Add](#)

Harvesters [Add](#)

INVITATIONS

Invitations [Add](#)

MANAGEMENT COMMANDS OVER HTTP

Management command jobs [Add](#)

Change group profile

[HISTORY](#) [VIEW ON SITE](#)

Soil Data Managers

Title:

Slug:

Logo: [Scegli file](#) Nessun file selezionato

Description:

Email:

Email used to contact one or all group members, such as a mailing list, shared email, or exchange group.

Keywords:

A space or comma-separated list of keywords

Access:

Public: Any registered user can view and join a public group.
Public (invite-only): Any registered user can view the group. Only invited users can join.
Private: Registered users cannot see any details about the group, including membership. Only invited users can join.

Categories:

[SAVE](#) [Save and add another](#) [Save and continue editing](#) [Delete](#)

Figure 11: group profile

[Home](#) > [Authentication and Authorization](#) > [Groups](#) > data-managers

Start typing to filter...

ACCOUNTS

Email addresses + Add

ACTIVITY STREAMS

Actions + Add

Follows + Add

ANNOUNCEMENTS

Announcements + Add

Dismissals + Add

ASSETS

Local assets + Add

AUTHENTICATION AND AUTHORIZATION

Groups + Add

AVATAR

Avatars + Add

BACKOFFICE

Change group

data-managers

NAME:

data-managers

Permissions:

Available permissions

Choose permissions by selecting them and then select the "Choose" arrow button.

Filter

Accounts | email address | Can add email address
Accounts | email address | Can change email address
Accounts | email address | Can delete email address
Accounts | email address | Can view email address

Chosen permissions

Remove permissions by selecting them and then select the "Remove" arrow button.

Filter

Backoffice | base dataset | Can add base dataset
Backoffice | base dataset | Can change base dataset
Backoffice | base dataset | Can delete base dataset
Backoffice | base dataset | Can view base dataset

Choose all permissions

Remove all permissions

Hold down "Control", or "Command" on a Mac, to select more than one.

SAVE

Save and add another

Save and continue editing

Figure 12: group permissions

SOIL DATA

The first action to perform after the installation is the “base dataset” configuration that can be performed in the BackOffice (see “Configure” in the Data Managers Manual)

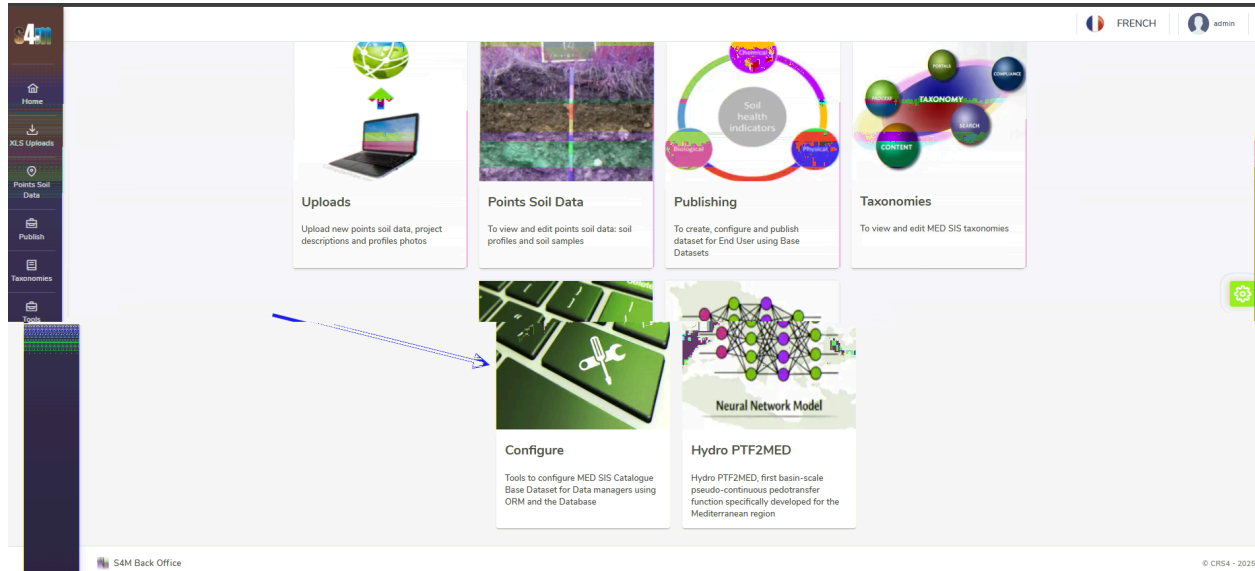


Figure 13: Back Office - Configure tool

It is recommended to use the catalog for managing datasets, but certain operations can be performed in the administration panel, as shown in Figure 14.

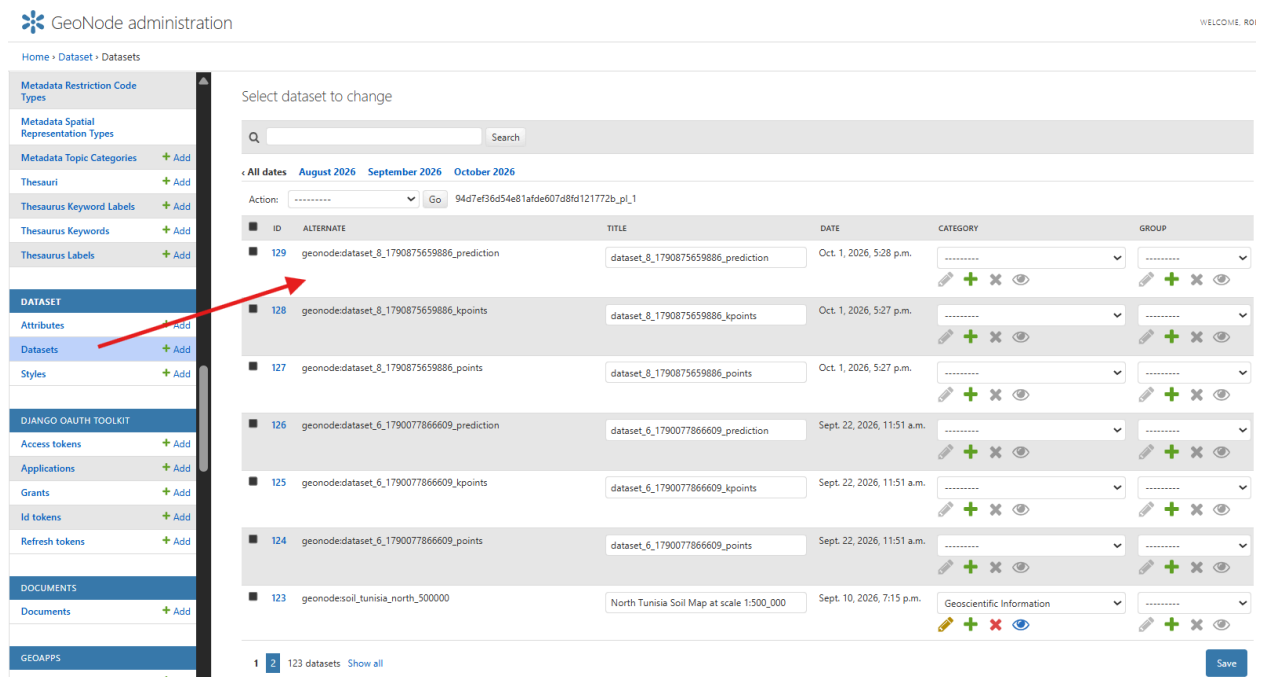


Figure 14: Admin Panel - Dataset Change

4. Backup and Restore

The backup and restore guide is available in the github repository at https://github.com/crs4/med_sis/blob/main/BACKUP_RESTORE_GUIDE.md